Cambridge (CIE) A Level Maths: Pure 3

Your notes

Further Trigonometric Equations

Contents

- Strategy for Further Trigonometric Equations

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Your notes

Strategy for further trigonometric equations

How do I solve harder trig equations?

- You can solve harder trig equations, such as those involving **reciprocal** and **inverse** functions in a variety of different ways
  - Using **further trigonometric identities**
  - Using **compound or double angle** formulas
  - Factorising **quadratic** trig equations
  - Then finding all solutions using **CAST** or **sketching graphs**
- The final rearranged equation you solve will involve **sin**, **cos** or **tan** – don't try to solve an equation with **cosec**, **sec**, or **cot** directly
- If you're having trouble solving a trig equation, this flowchart might help:

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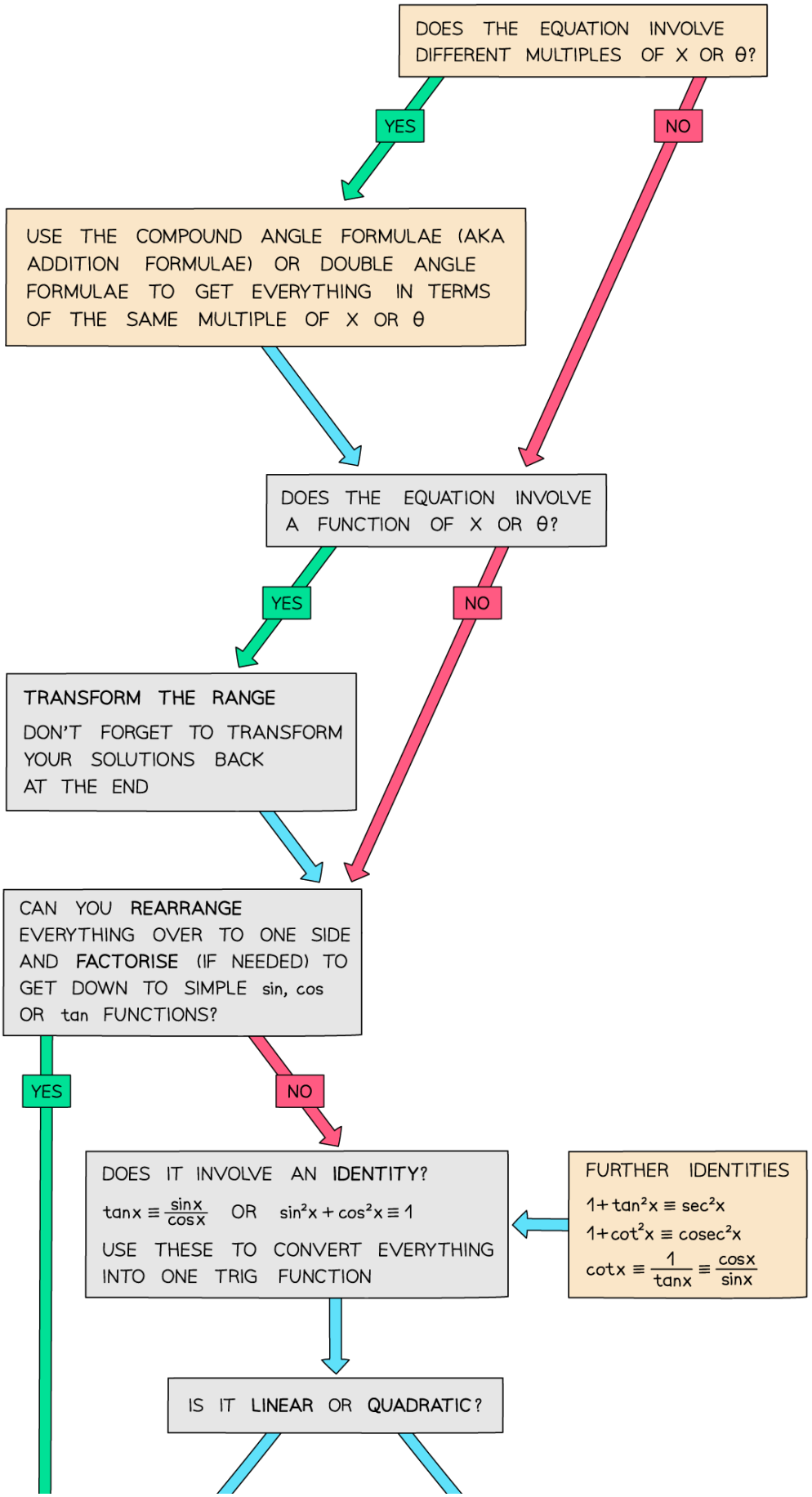
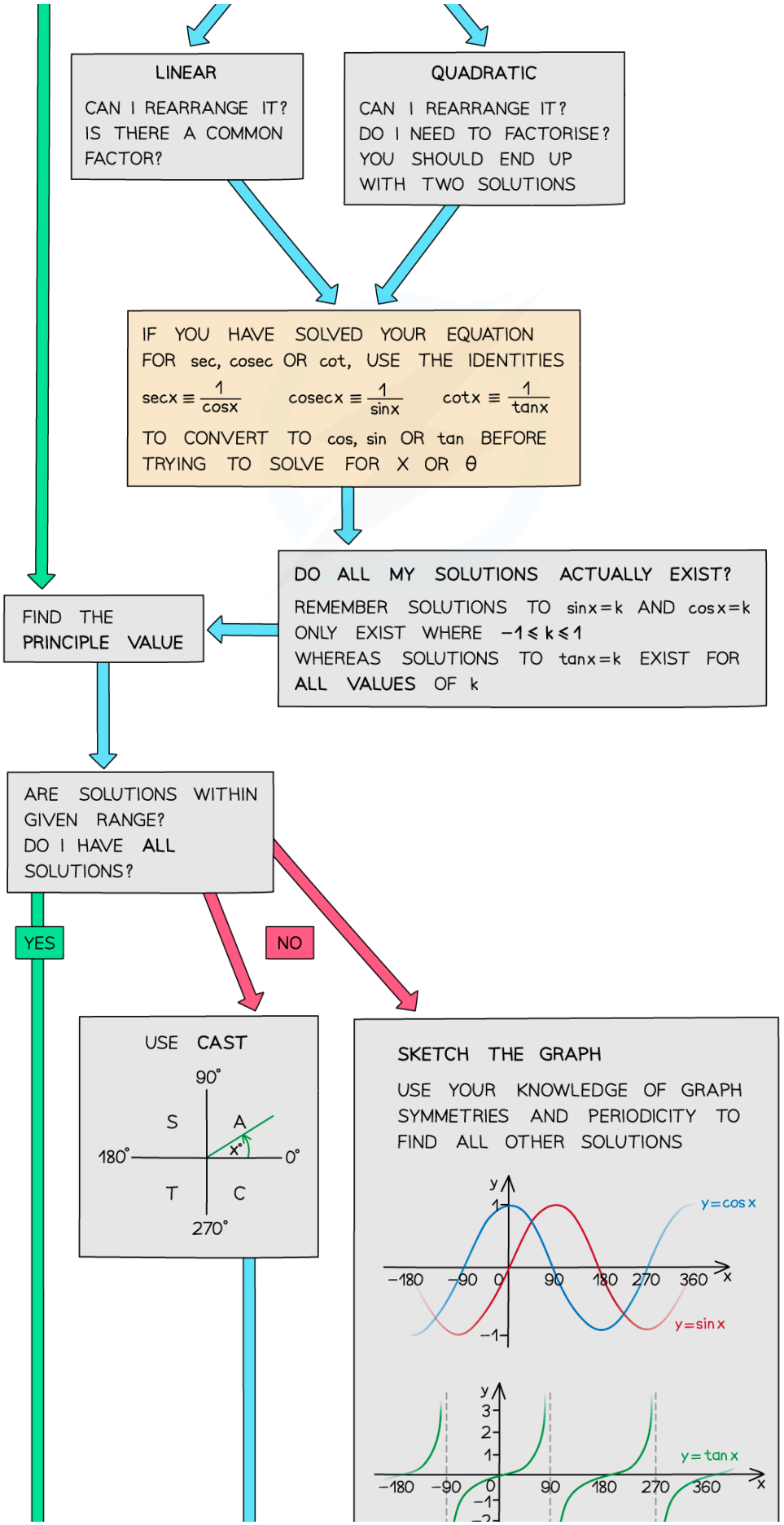

DOES THE EQUATION INVOLVE DIFFERENT MULTIPLES OF X OR θ?
YES
NO
USE THE COMPOUND ANGLE FORMULAE (AKA ADDITION FORMULAE) OR DOUBLE ANGLE FORMULAE TO GET EVERYTHING IN TERMS OF THE SAME MULTIPLE OF X OR θ
DOES THE EQUATION INVOLVE A FUNCTION OF X OR θ?
YES
NO
TRANSFORM THE RANGE
DON'T FORGET TO TRANSFORM YOUR SOLUTIONS BACK AT THE END
CAN YOU REARRANGE EVERYTHING OVER TO ONE SIDE AND FACTORISE (IF NEEDED) TO GET DOWN TO SIMPLE sin, cos OR tan FUNCTIONS?
YES
NO
DOES IT INVOLVE AN IDENTITY?
$\tan x \equiv \frac{\sin x}{\cos x}$ OR $\sin^2 x + \cos^2 x \equiv 1$
USE THESE TO CONVERT EVERYTHING INTO ONE TRIG FUNCTION
FURTHER IDENTITIES
$1 + \tan^2 x \equiv \sec^2 x$
$1 + \cot^2 x \equiv \operatorname{cosec}^2 x$
$\cot x \equiv \frac{1}{\tan x} \equiv \frac{\cos x}{\sin x}$
IS IT LINEAR OR QUADRATIC?

Your notes


LINEAR
CAN I REARRANGE IT?
IS THERE A COMMON FACTOR?
QUADRATIC
CAN I REARRANGE IT?
DO I NEED TO FACTORISE?
YOU SHOULD END UP WITH TWO SOLUTIONS
IF YOU HAVE SOLVED YOUR EQUATION FOR sec, cosec OR cot, USE THE IDENTITIES
$\sec x \equiv \frac{1}{\cos x}$ $\operatorname{cosec} x \equiv \frac{1}{\sin x}$ $\cot x \equiv \frac{1}{\tan x}$
TO CONVERT TO cos, sin OR tan BEFORE TRYING TO SOLVE FOR X OR θ
DO ALL MY SOLUTIONS ACTUALLY EXIST?
REMEMBER SOLUTIONS TO $\sin x = k$ AND $\cos x = k$ ONLY EXIST WHERE $-1 \leqslant k \leqslant 1$
WHEREAS SOLUTIONS TO $\tan x = k$ EXIST FOR ALL VALUES OF k
FIND THE PRINCIPLE VALUE
ARE SOLUTIONS WITHIN GIVEN RANGE?
DO I HAVE ALL SOLUTIONS?
YES
NO
USE CAST
90°
S
A
x°
180°
0°
T
C
270°
SKETCH THE GRAPH
USE YOUR KNOWLEDGE OF GRAPH SYMMETRIES AND PERIODICITY TO FIND ALL OTHER SOLUTIONS
y
1
−1
−180
−90
0
90
180
270
360
x
y = cos x
y = sin x
y
3
2
1
−1
−2
−180
−90
0
90
180
270
360
x
y = tan x

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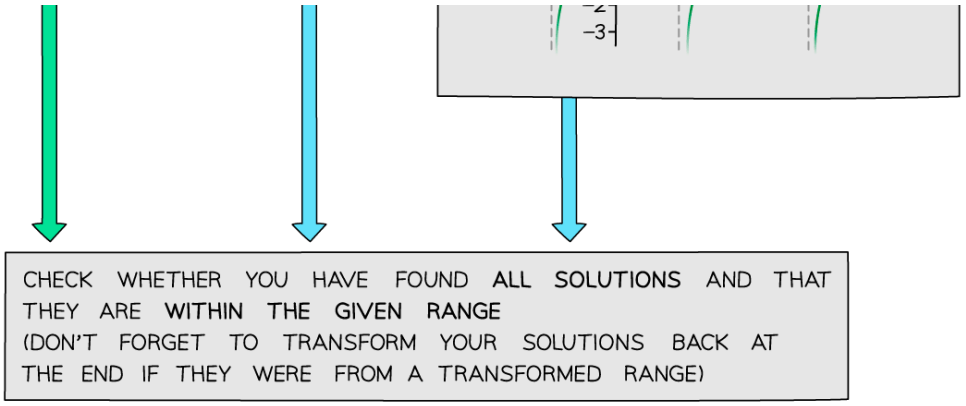

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Examiner Tips and Tricks

- Try to use identities and formulas to reduce the equation into its simplest terms.
- Don't forget to check the function range and ensure you have included all possible solutions.
- If the question involves a function of x or θ ensure you transform the range first (and ensure you transform your solutions back again at the end!).

Worked Example

Your notes

Find the solutions for the equation

$$\left(1+\cot^2 2\theta\right)\left(5\cos^2\theta - 1\right) = \cot^2 2\theta$$

in the interval $0 \le \theta < 360°$.

$$5\cos^2\theta - 1 = \frac{\cot^2 2\theta}{1+\cot^2 2\theta}$$

$$5\left(\tfrac{1}{2}\cos 2\theta + \tfrac{1}{2}\right) - 1 = \frac{\cot^2 2\theta}{\operatorname{cosec}^2 2\theta}$$

USE TRIG IDENTITIES AND FORMULAE

$1+\cot^2 A = \operatorname{cosec}^2 A$

$\cos 2A \equiv 2\cos^2 A - 1$ (DOUBLE ANGLE FORMULA)

$$\frac{1}{2}(5\cos 2\theta + 3) = \frac{\dfrac{\cos^2 2\theta}{\sin^2 2\theta}}{\dfrac{1}{\sin^2 2\theta}} = \cos^2 2\theta$$

$\cot A \equiv \dfrac{\cos A}{\sin A}$ $\quad \operatorname{cosec} A \equiv \dfrac{1}{\sin A}$

$$5\cos 2\theta + 3 = 2\cos^2 2\theta$$

$$2\cos^2 2\theta - 5\cos 2\theta - 3 = 0$$

THIS QUADRATIC IN $\cos 2\theta$ IS WHAT WE WILL ACTUALLY SOLVE!

$$(2\cos 2\theta + 1)(\cos 2\theta - 3) = 0$$

FACTORISE

SO $\cos 2\theta = -\frac{1}{2}$ OR $\cos 2\theta = 3$

BUT $\cos 2\theta = 3$ HAS NO SOLUTIONS

$-1 \leqslant \cos 2\theta \leqslant 1$

SO WE WANT SOLUTIONS TO $\cos 2\theta = -\frac{1}{2}$

IN THE RANGE $0 \leqslant 2\theta < 720°$

TRANSFORM THE RANGE

$2\theta = \cos^{-1}\left(-\frac{1}{2}\right) = 120°$

FIND PRINCIPAL VALUE

OR $2\theta = 360 - 120 = 240°$

USE CAST TO FIND OTHER VALUE IN RANGE $0 \leqslant 2\theta < 360°$

SO $2\theta = 120°, 240°, 480°, 600°$

ADD 360° TO FIND ADDITIONAL ANSWERS IN RANGE $360 \leqslant 2\theta < 720°$

$\theta = 60°, 120°, 240°, 300°$

CONVERT BACK TO θ

0 = 60, 120, 240, 300

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